

Quayle Companion Volume — Instructions for VECTOR (Charles reads this too)

Last synced: 2026-08-07

From: Charles Levine

To: VECTOR

Job: Guided reading companion for Quayle, Siddiqui & Jones (2006)

Charles's plan: Print this document, read it, upload **this file plus everything else in the Quayle folder** to VECTOR, then work Phase 1 → 2 → approval → 3.

0. What you are building (both of us)

A **companion volume** is **not** a summary, lit review, study guide, or paraphrase.

It is a guided intellectual edition: Charles reads the **source paper PDF side-by-side** with your commentary and feels like an attentive co-author is explaining what deserves underlining, what mechanisms matter, and what assumptions hide in the math.

Success test: *Reading beside an exceptionally insightful co-author.*

Failure modes: invented anchors, sparse quotes, executive-summary tone, placeholder chapters, excessive whitespace. See `Companion_Volume_Lessons_Learned.md`.

Menger precedent (June 2026): You and Charles built `Complete_Companion.pdf` from chapter Word `.docx` files (see `Menger_Companion_Chapter_10_EXAMPLE.docx` in this folder). **Quayle uses the same intellectual format** but you deliver **markdown** (`.md`); Charles converts to PDF locally.

1. Source paper

Quayle, A. P., Siddiqui, A. S., & Jones, S. J. M. (2006).
Modeling network growth with assortative mixing.
European Physical Journal B, 50, 617–630.

File in this folder: `Quayle_Siddiqui_Jones_2006_assortative_mixing.pdf` (~14 pages)

2. Files in this folder — read before you write anything

#	File	Role
1	<code>VECTOR_Quayle_Companion_Instructions.md</code>	This document — process, phases, scope
2	<code>Quayle_Siddiqui_Jones_2006_assortative_mixing.pdf</code>	Source manuscript
3	<code>Companion_Volume_Creation_Protocol_v2.md</code>	Binding operating manual
4	<code>Companion_Volume_Lessons_Learned.md</code>	What worked / failed on Menger
5	<code>Reader_Context_Profile.md</code>	How Charles reads (dense, quotes, print)
6	<code>Research_Context_Overview.md</code>	Background for light RED hooks only
7	<code>Complete_Companion.pdf</code>	Merged Menger companion — print density & structure target
8	<code>Menger_Companion_Chapter_10_EXAMPLE.docx</code>	One chapter exemplar — section skeleton, voice mix (Word era)
9	<code>Companion_Volume_Formatting_Guide.docx</code>	Original BLACK / BLUE / RED rules (Word)
10	<code>Companion_Markdown_Formatting_Guide.md</code>	How to encode the three voices in <code>.md</code> (HTML spans, unit template)
11	<code>Markdown_to_PDF_Formatting_Guide.md</code>	Math + Pandoc — <code>\$...\$</code> , <code>\$\$...\$\$</code> , symbols
12	<code>Companion_Volume_Request_TEMPLATE.docx</code>	Generic kickoff template from Menger/Rosen round (reference)

Additional materials: Charles can supply more on request (other Menger chapters, deep-reading notes, papers). **Ask first** — do not assume you need them.

3. Workflow — four phases (do not skip)

Phase 1 — Read everything

Before proposing or writing companion prose, read carefully:

1. This instructions file
2. `Companion_Volume_Creation_Protocol_v2.md` (binding)
3. `Reader_Context_Profile.md` + `Companion_Volume_Lessons_Learned.md`
4. `Complete_Companion.pdf` — note density, anchors, quote load, synthesis
5. `Menger_Companion_Chapter_10_EXAMPLE.docx` — one chapter's section structure
6. `Companion_Markdown_Formatting_Guide.md` + `Markdown_to_PDF_Formatting_Guide.md` — how Quayle deliverable must look in `.md`

7. `Quayle_Siddiqui_Jones_2006_assortative_mixing.pdf` — the full paper

Treat Menger as **style exemplar**; treat the protocol as **binding**.

Phase 2 — Propose companion architecture (required before drafting)

Do NOT begin writing companion chapters yet.

Identify **conceptual units** in Quayle — **not** page boundaries, **not** paragraph-by-paragraph commentary.

Deliver to Charles a **Companion Architecture Proposal** containing:

A. Chapter / unit list

For each proposed unit (aim ~8–12 units for this 14-page paper, plus synthesis + appendix):

Field	Requirement
Unit number	1, 2, 3, ...
Proposed title	Conceptual (e.g. “Scalar properties and the similarity kernel”)
Anchor	Verbatim opening text of the first paragraph in this block — never paraphrase, never invent
Major concepts	Bullet list
Relative importance	High / medium / low for Charles’s understanding

Suggested unit map (starting point — revise after reading):

1. Introduction — homophily vs assortativity, why generic vertex properties
2. Model core — Eq. (4), degree PA × similarity product
3. Discrete properties — δ function, Eq. (8), appendix $r(\alpha)$
4. Hierarchical properties — tree similarity, Eq. (12)–(14)
5. Scalar / continuous properties — **Eq. (15)–(16)** (high priority)
6. Community structure & modularity — what claims matter
7. Simulation results — discrete case (Figs. 2–4)
8. Simulation results — hierarchical case (Figs. 6–9)
9. Summary & limitations in the paper
10. **Final synthesis** (your voice — not a summary)
11. **Appendix** — one cross-cutting mechanism (e.g. homophily → assortativity → community)

B. Deep-structure memo

Also provide:

- Major **mechanisms** (phenomenon → mechanism → assumptions → boundaries)
- Recurring **themes**
- **Hidden assumptions**
- **Transferable concepts** (network science, not Charles’s code)
- Planned **synthesis chapter** focus
- Planned **appendix** focus

C. Formatting acknowledgment

Confirm you will use:

- `Companion_Markdown_Formatting_Guide.md` for unit skeleton and three voices
- `Markdown_to_PDF_Formatting_Guide.md` for all math

Phase 3 — Wait for Charles’s approval

After Phase 2, **STOP**.

Do not draft integrated commentary until Charles approves the architecture (he may rename units, merge/split, or reprioritize).

Phase 4 — Write the companion (after approval)

Deliver **markdown** (`.md`):

- One file `Quayle_Companion_Volume.md` preferred; or one `.md` per unit if size forces it (ask Charles).
- Charles converts: `./scripts/convert_single_md_to_pdf.sh Quayle_Companion_Volume.md Quayle_Companion_Volume.pdf pdf_styles_narrow.css`
- **You do not produce PDF.**

Unit template (each conceptual unit)

```
## Unit N - [Title]

### Anchor (Original Text)

<span style="color:#111;font-weight:bold">[/Verbatim opening sentence(s) from paper...]/</span>

### Key Quotations Worth Underlining

- **"[Quote 1]"**
- **"[Quote 2]"**

### Integrated Commentary

<span style="color:#1a5490">[/BLUE - what the author is really saying; usually longer than the source passage.]/</span>

<span style="color:#b03030">[/RED - optional; sparse; why this might matter beyond the paper.]/</span>

### Margin Notes Worth Scribbling in the Manuscript

- ...
```

Three voices:

Voice	Markdown	Job
BLACK bold	<code></code> or <code>**quote**</code>	Verbatim underline candidates
BLUE	<code>...</code>	Interpretation — dominant
RED	<code>...</code>	Implications — sparse ; synthesis + appendix mainly

Interweave BLUE and RED naturally — do not dump all RED at the end.

Synthesis chapter (required)

Answer (Protocol Part XIII):

- Why does this paper matter?
- Deepest contributions?
- What transfers?
- What assumptions deserve skepticism?
- What should Charles remember in five years?

Not a bullet summary of units.

Appendix (required)

One cross-cutting mechanism distributed through the paper (Protocol Part XIV).

Example title: *Homophily* → *assortativity* → *community structure*.

4. Quayle-specific reading priorities

Charles cares most about understanding:

- **Homophily** α vs **assortativity** r — vocabulary discipline
- **Governing equation (4)** — product of degree attachment and similarity preference
- **Scalar properties (§2.3, Eqs. 15–16)** — similarity kernel and attachment law (**high priority**)
- Discrete vs hierarchical cases — when each matters
- **Appendix** — ensemble $r(\alpha)$ and inverting for α (Eq. 28)
- Community / modularity results — main claims vs detail Charles can skim

5. RED / project connections — light touch

This companion is for **paper mastery**, not simulation implementation.

- Interweave RED **sparingly** in unit commentary.
- A **few paragraphs total** in synthesis + appendix may note: Charles is reading this because **assortative mixing** is intellectually adjacent to **assortative matching** in organizational/advancement work (`Research_Context_Overview.md`).
- **No** file paths, function names, code, or “assign rule” deliverable.

6. Quality bar (checklist before handoff)

- ☐ Architecture was **approved** before drafting
- ☐ Every unit has **verbatim Anchor** — underline test passed

- ☐ **Key Quotations** in every unit — dense, not sparse
- ☐ BLUE generally **longer** than the passage it interprets
- ☐ All math in `$...$` or `$$...$$` per formatting guide
- ☐ **Last synced:** date under document `#` title
- ☐ Synthesis + appendix present — not placeholders
- ☐ Document converts cleanly for Charles (no empty `\[ \]`, no bare LaTeX lines)

7. What Charles does after you deliver

1. Review `.md`
2. Run PDF conversion on his Mac
3. Print and read beside `Quayle_Siddiqui_Jones_2006_assortative_mixing.pdf`

8. Start here (VECTOR)

Your first deliverable is Phase 2 only: Companion Architecture Proposal + deep-structure memo.

Do not write Unit 1 until Charles approves.

When ready, send the architecture proposal to Charles in chat.

End of instructions.